AWS Static Website Deployment with S3 and CloudFront

Overview

This setup provisions an **S3 bucket** for hosting a static website and configures **CloudFront** to serve the site with improved **performance, security, and global availability**.

S3 Bucket

- Hosts the static website
- Bucket Name: ani-static-website-197
- Website Endpoint: http://ani-static-website-197.s3-website-us-east-1.amazonaws.com
- Public-read ACL enabled
- Static website hosting configured

Bucket Policy (Public Read)

```
{
"Version": "2012-10-17",
"Statement": [
{
"Sid": "PublicRead",
"Effect": "Allow",
"Principal": "*",
"Action": "s3:GetObject",
"Resource": "arn:aws:s3:::ani-static-website-197/*"
}
]
}
```

CloudFront CDN

1. Delivers content via **global CDN**
2. Distribution Domain Name: d2iz4xijy4o0i5.cloudfront.net
3. ARN: arn:aws:cloudfront::247842832518:distribution/E2AYGZTZR65EE6
4. Viewer Protocol Policy: **Redirect HTTP to HTTPS**
5. Default Root Object: index.html
6. Caching enabled for performance optimization

Manual Implementation

Steps to manually set up the static website and CloudFront distribution.

1. Create an S3 bucket and enable static website hosting.
2. Set the bucket policy to allow public read access.
3. Configure a CloudFront distribution with the S3 bucket as the origin.
4. Enable HTTPS and set default caching settings.

Terraform Implementation

1. Automates all manual steps
2. Code provisions S3 bucket, bucket policy, and CloudFront distribution
3. Variables like bucket name are configurable
4. Outputs:

Output	Description
bucket_website_endpoint	S3 static website endpoint
cloudfront_url	CloudFront distribution URL
cloudfront_arn	CloudFront distribution ARN

Infrastructure Configuration

1. S3 Bucket : Public-read ACL; Static website hosting
2. Bucket Policy : Public read access policy
3. CloudFront Distribution : HTTPS via CloudFront; Caching for better performance

Folder Structure

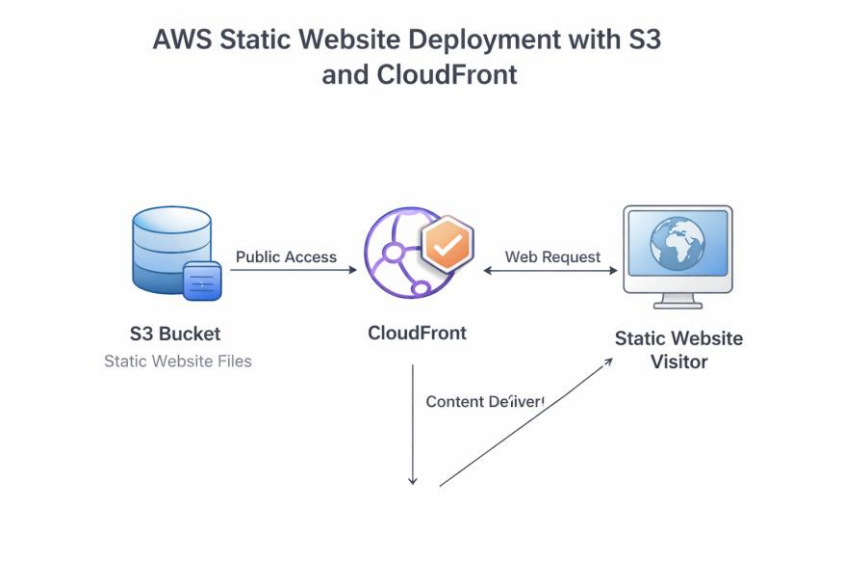
```
Project_2/
    ├── 01_Static_website/     # index.html, error.html
    ├── 02_terraform/          # main.tf, variables.tf, outputs.tf
    ├── 03_Documentation/      # screenshots, architecture diagram
    └── README.md
```

Architecture Diagram

Flow:

Browser → CloudFront → S3 Bucket → index.html / error.html

1. CloudFront caches content globally
2. HTTPS enforced via CloudFront
3. S3 serves static content



Key Learnings

1. Hosting static websites using S3
2. Managing public access using bucket policies
3. Using CloudFront as CDN
4. Enabling HTTPS without custom domain
5. Understanding origin and caching behaviour

Future Work

1. Custom Domain : Integrate a custom domain with HTTPS
2. Logging & Monitoring : Enable logging and monitoring for analytics
3. Security Enhancements : Implement advanced security measures

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